Denoising low-intensity diffraction signals using *k*-space deep learning: Applications to phase recovery

Sung Yun Lee,[1,2] Do Hyung Cho,[1,2] Chulho Jung,[1,2] Daeho Sung,[1,2] Daewoong Nam,[2,3] Sangsoo Kim,[2,3] and Changyong Song[1,2,*]

[1]*Department of Physics, POSTECH, Pohang 37673, Korea*
[2]*Photon Science Center, POSTECH, Pohang 37673, Korea*
[3]*Pohang Accelerator Laboratory, POSTECH, Pohang 37673, Korea*

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Phase recovery is a well-known inverse problem prevalent across science disciplines and attracts active research interests to develop a number of theoretical and experimental methods. Recent developments in artificial intelligence have further prompted research activities in processing the experimentally collected imperfect data, but applications have been limited to slow-varying data such as real images. Experimental noise present in largely fluctuating diffraction data, in particular, adds practical challenges to hamper consistent phase recovery. Here, we introduce a convolutional neural-network assisted *k*-space denoising method that can directly manage noisy diffraction signals. It showed superior performance on denoising the diffraction data, which promote improved phase recovery from noise-buried single-pulse diffraction signals obtained by the x-ray free-electron laser. Adapting our method to general diffraction data can expand boundaries of interpretable data and enhance observability of faint objects with weak signals.

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Recent progress in both digital technologies and computational algorithms has opened up new avenues in data science [1–3]. Scientific research relies on data collection and processing and developing effective as well as accurate methods is essential to extract the most useful information from the available data. Imaging science inherits this quest for smart data handling, which is indispensable, considering the remarkably large data size [4]. Neural-network techniques developed in artificial intelligence research have reduced the need for human inspection with notable progress in noise deblurring, adaptive illuminations, holographic phase recovery, and ghost imaging [2,5,6].

A convolutional neural network (CNN) is a neural network in which layers have convolution operations with kernels instead of linear combinations [7]. It is an effective pattern recognition method to classify image data with significantly reduced computational complexity while still providing the major characteristics of the image data [8]. The CNN treats the input data with repeated operations of numerical convolutions that reduce the data size while preserving the main distinctive features of the data [9]. Recovering images taken at low photon counts was demonstrated by deep-learning assisted phase retrievals [10]. Applications in holographic imaging were also demonstrated with faster recovery of real images by overcoming complexities, such as twin images and self-interference artifacts [11]. However, machine-learning techniques for experimental data in *k* space, inverse length, or reciprocal space has not been developed actively, although nearly all the scattering data are in *k* space. In particular, exponential-scale intensity variations in diffraction data hamper causal data extrapolations. Interpretations of *k*-space data with the deep learning was demonstrated recently to conject probable real-space images of constrained homogeneous nanoparticle systems, but still the processing of noisy *k*-space data has not been realized [12–14].

In this paper, we implement a *k*-space CNN model to process the noise in coherent diffraction patterns to promote enhanced phase retrieval performance for single-pulse femtosecond diffraction patterns of the x-ray free-electron laser (XFEL) without any prior object information. By retrieving the phase information of the diffraction intensity collected using coherent wave sources, the object image was acquired via coherent diffraction imaging [15,16]. Iterative numerical operations of discrete Fourier and inverse Fourier transformations were performed to retrieve the phase buried in the oversampled diffraction data [15,17,18]. The phase retrieval was realized using only the far-field diffraction data and not supported by any holographic phase interference or object image. Imperfect experimental data with noise contamination or data loss complicate performing phase recovery. Many efforts have been made to introduce noise-robust phase retrieval algorithms by suppressing high-frequency noise or using a proximal mapping of an objective function with relaxed constraints [19,20]. Also, CNN-based modifications showed meaningful improvements by adjusting temporal real-space variables or applying real-space denoisers as a regularizer

*cysong@postech.ac.kr

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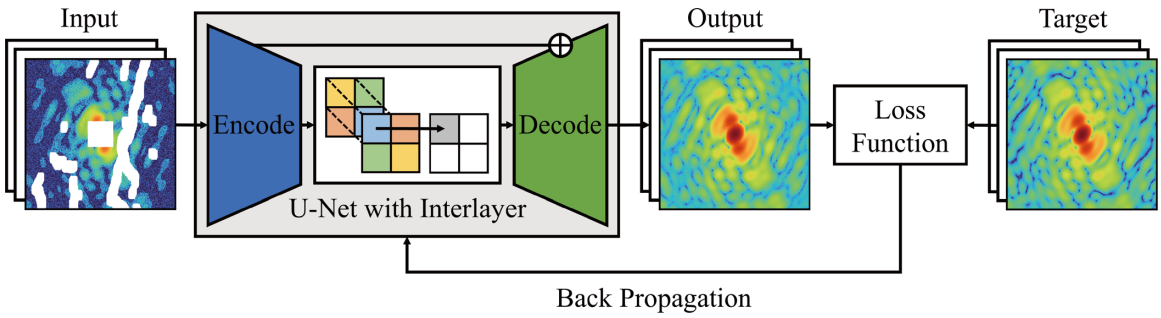

FIG. 1. Convolutional neural network for k-space deep denoising. Low- and high-intensity diffraction patterns were generated from random objects with Poisson noise and additive Gaussian noise. The low-intensity patterns were masked with random irregular patterns (input) to simulate experimental situations with partial data loss, and high-intensity patterns (target) are compared with the outputs of the network using the loss function. The network has a U-Net architecture in which the decode layers have skipped connections with corresponding encode layers, and convolution operations are replaced by partial convolutions. In the interlayer, the outputs from the encoder layers convolved with their centrosymmetric points one by one. The loss function considers both output patterns and autocorrelation of corresponding real-space objects. The parameters of the network were updated by the partial derivatives of the loss function propagating in the backward direction, which is called back-propagation.

[21,22]. However, as the noise becomes stronger, the performance of those approaches becomes poorer with low convergence. A better approach is to reduce the noise of the diffraction pattern itself. Our denoising method based on deep learning effectively reduces the noise, which promotes accurate phase retrievals by extending the tolerance to strong noise [11,18,23–27].

We introduce a k-space CNN model to denoise experimentally obtained diffraction patterns with strong Poisson noise (Fig. 1). The method employs U-Net-based CNN using partial convolutions with a centrosymmetric pointwise convolution interlayer [28,29]. U-Net is a type of CNN, which has an encoder-decoder structure with inputs of each decoder layer concatenated with the equal-sized outputs of corresponding encoder layers (i.e., skipped connections). The encoder layers have partial convolutions followed by normalization and nonlinear operations on the output variables for appropriate network training, while the decoder layers have upsampling operations via the nearest-neighbor interpolation and skipped connections before those computations [30]. Partial convolution is performed with a weight correction by ignoring lost values in convolved regions to avoid influences of the missing regions from a beam stopper and hardware arrangements of detector panels. The additional interlayer accounts for the centrosymmetry in the diffraction patterns, valid for diffractions from objects without notable absorptions, by pointwise convolutions on centrosymmetric point pairs. Inputs of the network were log scaled to handle exponential-scale intensity differences between low- and high spatial-frequency regions.

For network training, we generated a diffraction pattern dataset by calculating the discrete Fourier transform of pseudorandom images using the following equation:

$$I_i^{\text{obs}} = \text{Poisson}\left(|\mathcal{F}u|_i^2 \cdot \frac{I^{\text{sc}}}{\sum_j |\mathcal{F}u|_j^2}\right) + \mathcal{N}(0,\sigma), \qquad (1)$$

where u is a real-space object, $\mathcal{F}$ is a discrete Fourier transform operator, and I^{sc} is the total scattered intensity. I^{sc} was set to 2×10^6 photons for low-intensity patterns to emulate the XFEL single-pulse measurements and 100 times larger for high-intensity patterns. An additive Gaussian noise term, σ, was set to 0.25. For unbiased pseudorandom datasets to avoid overfitting while training, the objects were generated by the products of two random images from preexisting datasets: EMNIST and CIFAR-10 [31,32]. EMNIST, a handwriting dataset, provided object shapes, while CIFAR-10, a natural-image dataset, gave the density distribution of the objects. The obtained patterns were masked by NVIDIA irregular mask dataset and a randomly sized beam stopper positioned at the center [29].

With the generated data, we used a composite loss function to update network parameters [29]. It is defined as a linear combination of multiple loss functions: L1 loss of the output ($\mathcal{L}_{\text{full}}$) and filled missing regions ($\mathcal{L}_{\text{missing}}$), perceptual loss ($\mathcal{L}_{\text{percept}}$), style loss ($\mathcal{L}_{\text{style}}$), centrosymmetric loss ($\mathcal{L}_{\text{centro}}$), and total variation loss ($\mathcal{L}_{\text{tv}}$). Perceptual loss and style loss are L1 loss of high-level features and their autocorrelation, respectively. High-level features are defined as each output of the first to third layer in the ImageNet-pretrained VGG-16 network through which the input was passed [33,34]. Centrosymmetric loss is an L1 distance between the output and the flipped output, while total variation loss is the distance between filled missing regions of output images with 1-pixel dilation. All functions were linearly combined with specific coefficients defined as Eq. (2) to avoid unexpected artifacts shaped like a checkerboard or fish scales [29]. Using the combined function, the loss was calculated with the output patterns and autocorrelation (AC) of corresponding real-space objects derived by inverse Fourier transform of the diffraction intensity.

$$\begin{aligned}\mathcal{L}_{\text{total}} = {} & \left(\mathcal{L}_{\text{full}} + 0.1\mathcal{L}_{\text{full}}^{\text{AC}}\right) + 5\mathcal{L}_{\text{missing}} \\ & + 0.05\left(\mathcal{L}_{\text{percept}} + 0.1\mathcal{L}_{\text{percept}}^{\text{AC}}\right) \\ & + 120\left(\mathcal{L}_{\text{style}} + 0.1\mathcal{L}_{\text{style}}^{\text{AC}}\right) + 0.1(\mathcal{L}_{\text{centro}} + \mathcal{L}_{\text{tv}}) \qquad (2)\end{aligned}$$

ADAMW optimizer was used for a backpropagation of the loss metric with a gradually decreasing step size in the range

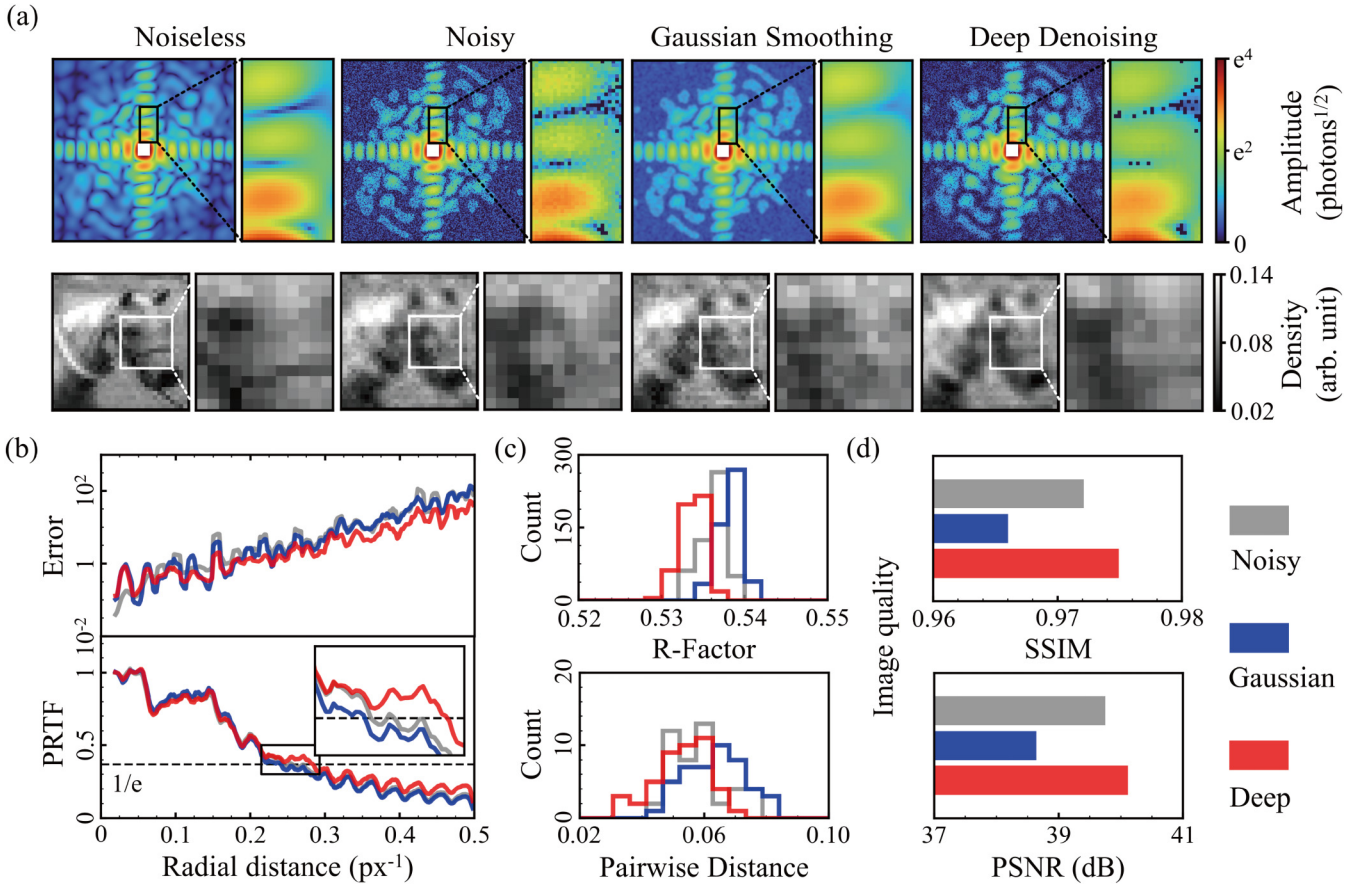

FIG. 2. k-space deep denoising verified using simulated data. (a) Diffraction patterns (top) and corresponding reconstructed objects (bottom). The ideal diffraction pattern (noiseless) was obtained from the Fourier transform of the test image, and the noisy pattern (noisy) was generated by adding Poisson and Gaussian noise to the noiseless pattern. The noisy pattern was convolved with a Gaussian kernel (Gaussian smoothing) to generate a pattern for comparison with the k-space deep denoised diffraction pattern employing the deep denoising network (deep denoising). The denoising method was further evaluated by inspecting the reconstructed images. The zoomed-in images indicated improved image quality for the deep denoised data. (b) Quantitative estimation of a radial error (top) and the PRTF (bottom) showed improved performance from the deep denoised data (red) compared to the Gaussian smoothed (blue) and noisy data (gray). (c) Reconstruction convergence compared using the R factor (top) and pairwise distance (bottom). (d) Bar charts of SSIM (top) and PSNR (bottom) comparing the image quality of the reconstructed objects demonstrated improved performance for the deep denoised data.

of 0.01–0.000 01 [35]. The direct denoising of the diffraction pattern by the trained network is hereinafter referred to as deep denoising.

We have demonstrated the performance of the deep denoising method via the CNN for the simulated diffraction patterns. The noisy diffraction patterns were generated by Eq. (1) using a test image. To evaluate the performance of deep denoising, we compared the Gaussian smoothed data by convolving a Gaussian kernel to the noisy pattern. Gaussian smoothing effectively reduced the noise in the major fringes but misguided the data points between speckles and the high spatial-frequency regions, leading to high noise in the low-intensity regions. In contrast, deep denoising recovered the speckle contrast with excellent performance while handling the noise near dips between the speckles [top panel in Fig. 2(a)]. The deep denoised data displayed reduced error, defined as a radial average of absolute difference with respect to the noiseless pattern, compared to others [top panel in Fig. 2(b)]. These results showed that the k-space deep denoising method recovers diffraction data more accurately.

The improved quality of the deep denoised data was further verified by performing phase retrievals to recover the image. Numerical phase retrieval was then performed by employing the relaxed-averaged alternating reflection (RAAR) method [36]. The RAAR method relies on the weighted average of the Douglas-Rachford projection operator and the projection operator on the k-space constraint [37]. Here, we modified the constraint of its latter projection operator with the denoised pattern in our model [38]. The use of a deep denoised pattern expedited the projection closer to the noiseless pattern during the iterations, improving the accuracy of phase recovery. The R factor, $R_F(u) = \sum_i ||\mathcal{F}u|_i - \sqrt{I_i^{\rm obs}}|/\sum_i \sqrt{I_i^{\rm obs}}$, was used for the error metric of the phase retrieval. Ten trials of 50 independent phase retrievals, a total of 500 independent phase retrievals, were performed to obtain the final reconstructions with the lowest error values. Ten images with the lowest error

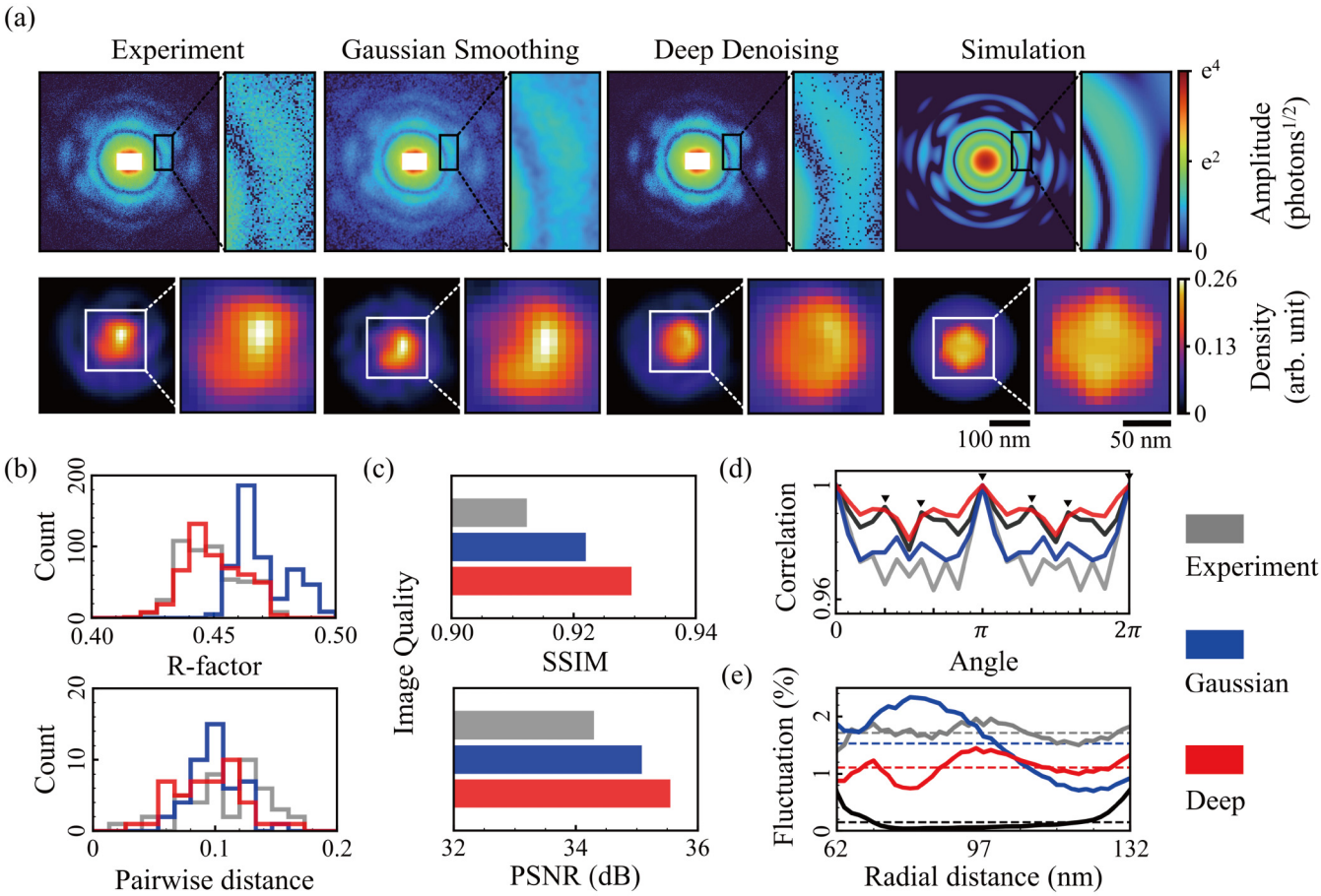

FIG. 3. k-space deep denoising on single-pulse XFEL diffraction patterns from a single core-shell nanoparticle. (a) Coherent diffraction patterns (top) and corresponding reconstructed images (bottom). Measured single-pulse pattern (experiment) was convolved with a Gaussian kernel (Gaussian smoothing) for comparison with the k-space deep denoised data (deep denoising). A simulated pattern was constructed from the nominal core-shell structure with a TOH Au core and a spherical TiO_2 shell as a reference (simulation). Improved denoising from the k-space deep denoising along with a more accurately reconstructed image demonstrates the successful application of deep denoising for experimental diffraction data. The zoomed-in views of the selected area are displayed for comparison. (b) Convergence of the phase retrieval was estimated by the R factor (top) and pairwise distance (bottom). (c) Quantitative assessment of the reconstructed object by comparing the SSIM (top) and PSNR (bottom). (d) The angular correlation plot manifested the sixfold symmetry resolved in the reconstructed image from the deep denoised data. (e) Significantly reduced fluctuation in the shell density for deep denoised data (red) was noted compared to the experimental (gray) and Gaussian smoothed data (blue). The level of fluctuation for the ideal simulated data (black) was provided as a reference. The broken lines guided the average level of fluctuations for each denoising scheme.

values were obtained from each set of 50 reconstructions. Then, the performance of the k-space deep denoising was assessed using the reconstructed images. The image resolution and convergence of the phase retrieval results were evaluated by the phase retrieval transfer function (PRTF) [39]. The PRTF estimation showed that the reconstruction of the deep denoising data showed improved phase retrieval with a higher resolution [Fig. 2(b), bottom]. Also, we checked the consistency noted among the reconstructed images by estimating the pairwise distance, $d_{\text{pair}}(u, u') = \sum_i |u - u'|_i / \sum_i |u + u'|_i$, for the best ten images, which showed better convergence among the reconstructed images without compromising the R-factor distribution [Fig. 2(c)]. Further, the structural similarity index measure (SSIM) and the peak signal-to-noise ratio (PSNR) were introduced to assess the quality of the real-space objects with respect to the ideal objects without the noise [40]. Both metrics measure the distance between two images, but the SSIM considers a perceptual similarity with luminance, structure, and contrast. The results from the deep denoised data showed better SSIM and PSNR with respect to the ideal object [Fig. 2(d)].

Overall, the reconstruction from the deep denoised diffraction data showed improved image quality, as supported by quantitative measures and the image itself displaying the detailed features [bottom panel in Fig. 2(a)]. As for the Gaussian smoothed data, the increase in low photon levels led to large fluctuations among the reconstructed images, which did not converge properly. The analysis of the simulation data confirmed that the phase retrieval of the k-space deep denoised data showed more accurate image reconstructions with enhanced consistency.

Next, we demonstrated the k-space deep denoising method applied to femtosecond x-ray diffraction patterns collected using the XFEL. With the renovating performance of the XFEL in generating the data in few thousands to hundreds of million times enhanced rate compared to the high-end synchrotron

x-ray sources, the application of the machine-learning technique to the XFEL single-pulse data is in significant demand. The inspection of individual data files becomes a significant bottleneck for the processing of such massive data, and automatic data processing with reliable performance on the noisy data is particularly critical for the strong shot noise in the single-pulse XFEL data [23]. For verification, we collected the single-pulse XFEL diffraction patterns of a single core-shell nanoparticle consisting of a trisoctahedral (TOH) Au core and a spherical TiO_2 shell. The experiments were performed using the XFEL at the Pohang Accelerator Laboratory [27]. Femtosecond x-ray pulses of 5-keV photon energy were focused by a pair of KB mirrors to obtain a 5-μm (horizontal) by 7-μm (vertical) radiation spot at the focal point with an average number of photons in a single pulse of 10^9 photons μm^{-2} [41].

We also compared the measured diffraction patterns with the calculated diffraction pattern from a nominal three-dimensional structure model with a TOH core particle capped with a spherical shell to evaluate the reliability of the experimental data. We inspected the first dominant speckle, next to the zeroth fringe of a circular shape, which depicts the morphology of the core particle. The deep denoised experimental data showed reduced noise level while maintaining the shape of the major speckles located at the sixfold symmetric position alongside the first fringe with a closer resemblance to the calculated pattern compared to the others. The Gaussian smoothing method handled the noise on the major speckles but increased the noise level in the high-frequency region while averaging over the shot noise [top panel in Fig. 3(a)]. The deep denoising showed better performance by effectively suppressing the noise.

Phase retrievals of the measured diffraction patterns with and without k-space deep denoising were performed to compare their imaging performance. The deep denoised data showed improvement but at a similar level to the experimental data. However, notable improvement was obtained for the R-factor values with lower values overall compared to the Gaussian smoothed data [Fig. 3(b), top]. Both deep denoised and Gaussian smoothed data showed a similar level of convergence, estimated by the pairwise distance with improved performance than as-obtained experimental data [Fig. 3(b), bottom]. In the SSIM and PSNR analysis, both deep denoised and Gaussian smoothed data improved compared to the experimental data, but the phase retrieval with the deep denoised pattern outperformed the phase retrieval with the Gaussian smoothed pattern [Fig. 3(c)].

The angular correlation of the projected shape of the TOH core and the radial fluctuation of the shell density were further calculated for the detailed structural analysis of a core-shell nanoparticle. The angular correlation reflects the rotational symmetry of the core shape. The radial fluctuation, defined as the standard error divided by the average of each radial region, evaluates the reliable reconstruction of the shell structure with a homogeneous density profile. The comparison evidenced that only the reconstruction from the deep denoised diffraction pattern recovered the sixfold symmetry in the angular correlation analysis, which is consistent with the hexagonal symmetry of the projected shape of the TOH core [Fig. 3(d)]. The density fluctuation of the shell region was estimated to be much lower for the reconstructed image using the deep denoised data [Fig. 3(e)]. Furthermore, deep denoised data only successfully recovered two-peak structures from the TOH structure in its internal density [bottom panel in Fig. 3(a)]. These results demonstrated that the k-space deep denoising method significantly improved phase retrieval accuracy obtained from the experimental diffraction patterns with strong shot noise.

Here, we introduced a deep-learning based denoising method directly applicable to the data with exponential-scale signal fluctuations and partially damaged data. The quantitatively improved performance due to accurate phase retrievals with the denoising method from shot-noise dominated single-pulse femtosecond x-ray diffraction patterns was demonstrated. The application of the proposed k-space denoising method is not exclusive to the diffraction data but can be extended to highly fluctuating data. Facilitated by automatized rapid handling of massive datasets, our method can impact single-pulse dynamic imaging using femtosecond XFEL pulses in which the data collection is realized using single-pulse patterns with strong shot noise with the successful recovery of the diffraction signals buried by the noise [26,27,42,43]. We believe that the demonstrated k-space deep denoising method will enable faint signals to overcome strong random noise.

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C.S. conceived the project; S.Y.L. developed the k-space deep denoising method and performed the simulation and data analysis. All contributed to the XFEL experiment. C.S. and S.Y.L. wrote the manuscript with input from all authors.

The authors declare no competing interests.

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